

ADVANCED SQL DATE FUNCTIONS Exercise 2.2.

1. SELECT emp_id, name, hire_date,
DATEADD(month, 6, hire_date) AS hire_date
FROM Employees

emp_id	name	hire_date	hire_date + 6 months
1	Alic	2025-01-15	2026-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

2. SELECT student_id,
name;
DATEDIFF(day, dob, GETDATE()) AS age_in_days;
FROM Students;

student_id	name	age_in_days
101	Maya	7354
102	Ethan	7806
103	Senna	7337

Q3. SELECT event_id,
event_name,
DATEDIFF(day, GETDATE(), event_date) AS
days_remaining

event_id	event_name	days_remaining
1	Seminar	-671
2	Workshop	-228
3	Hackathon	-452

Q4. SELECT invoice_id,
issue_date,
due_date,
DATEDIFF(day, issue_date, due_date) AS
days_between
FROM Invoices;

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-03-15	14
503	2025-04-10	2025-04-20	10

Q5. SELECT course_id,
name,
TO_CHAR(start_date, 'Month JJJ') AS format_date
FROM Courses;

course_id	name	format_date
201	Algebra	May 2025
202	Python	June 2025

Q6. SELECT member_id,
DATE_FROM_PARTS(start_year, start_month, start_date)
AS full_start_date
FROM Memberships;

member_id	full_start_date
1	2023-05-10
2	2022-11-25

Q7. SELECT sub-id,
plan,

DATEADD(year, 1, renewal_date) AS extend_renewal_date
FROM Subscriptions

sub-id	plan	renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Q8. SELECT order-id,
order_date,

CURRENT_DATE AS today_date,

DATEDIFF(day, order_date, CURRENT_DATE)

AS days_since_order

FROM Orders;

order-id	order_date	today_date	days-since-order
1001	2025-04-15	2026-04-16	366
1002	2025-04-10	2026-04-16	371

Q9 SELECT training-id,
topic,

EXTRACT(YEAR FROM training_date) AS training_year

FROM Trainings;

training-id	topic	training_year
1	Safety	2025
2	Compliance	2025

10. SELECT post-id,
title,

EXTRACT(hour FROM published_on) AS hr_pub,
EXTRACT(MINUTE FROM published_on) AS min
FROM Blog-posts;

post-id	title	hr-pub	min-pub
1	SQL Tips	10	15
2	Data Cleaning	16	45

Q11 SELECT driver-id,

license_expiry,

DATEDIFF(day, CURRENT_DATE, license_expiry) AS days-left

FROM Drivers;

driver-id	license_expiry	days-left
301	2025-08-10	-249
302	2028-12-31	-837

Q12.

SELECT message-id,

sent_timestamp,

CURRENT_TIMESTAMP AS current_timestamp,

DATEDIFF(second, sent_timestamp, CURRENT_TIMESTAMP)

AS seconds-since-sent

FROM Messages;

~~message-id~~ sent

13. SELECT return_id,
return_date,
DATEADD(day, 15, return_date) AS restock_date
FROM Returns;

14. SELECT assign_id,
TO_DATE('assigned_on', 'YYYY-MM-DD') AS assigned_on_date
FROM Assignments.

15. SELECT meeting_id,
TO_CHAR(scheduled_time, 'Month DD, YYYY "at" HH:MI AM') AS formatted_meeting_time
FROM Meetings;

meeting_id	formatted_meeting_time
1	April 29, 2022 at 02:00 PM
2	April 19, 2022 at 09:30 AM